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Consortium Blockchain for State-Owned Asset Transaction Polycentric Governance: An Evolutionary Game-Theoretic and Agent-Based Modeling Framework

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Abstract—The governance of state-owned asset (SOA) transactions in China faces persistent structural challenges rooted in principal-agent problems and information asymmetry. The prevailing "post-transaction audit" paradigm creates regulatory lag of six to eighteen months, during which non-compliant behaviors—including asset undervaluation, insider trading, and procedural circumvention—evade timely detection. This paper proposes a consortium blockchain-based governance framework that integrates smart contract automation with incentive-compatible mechanism design, and validates it through a dual-validation paradigm combining evolutionary game theory with agent-based modeling (ABM) using the Mesa library.

The game-theoretic analysis derives threshold functions $\lambda^* = f(B, C_e, p, D)$ and $p^* = g(B, C_e, \lambda, D)$ that characterize the penalty-detection substitution relationship. On-chain regimes ($p > 0.7$) require substantially lower penalty coefficients ($\lambda^* \in [0.15, 0.45]$) than off-chain regimes ($p < 0.4$, $\lambda^* \in [1.8, 3.2]$), reducing deterrence requirements by an order of magnitude. Mesa ABM simulations validate these predictions under bounded rationality and heterogeneous agent populations: parameter configurations exceeding threshold functions achieve 85.2% compliance convergence within 30 iterations (SD = 3.1%), compared to 27.8% under baseline off-chain conditions (SD = 6.4%), with the two populations diverging rapidly from a common 50% starting point. The system sustains over 80% compliance even under 20% Byzantine node infiltration, confirming collusion resistance at realistic adversarial levels. Sensitivity analysis reveals a "compliance ridge" along $\lambda \cdot p > 0.6$, demonstrating that penalty intensity and detection probability are policy substitutes amenable to regulatory engineering. These results represent conservative estimates under the current parameterization; more aggressive penalty-detection configurations yield higher compliance ceilings approaching 94%.

These findings establish that consortium blockchain does not merely improve transparency as a technological overlay, but fundamentally reshapes the strategic payoff landscape by directly controlling λ, p, and τ. The proposed framework provides concrete design targets—including a recommended penalty threshold of $\lambda \geq 2.5$ with inspection probability $p \geq 0.85$ for transactions exceeding ¥10 million—and a transferable dual-validation methodology for incentive mechanism design in multi-stakeholder governance contexts.

Index Terms—Consortium Blockchain, State-Owned Asset Transaction, Polycentric Governance, Evolutionary Game Theory, Agent-Based Modeling, Mesa Simulation, Incentive-Compatible Mechanism Design

1 Introduction

1.1 Research Background

State-owned assets (SOA) constitute a fundamental pillar of China's economic system, with total assets under central and local government management exceeding ¥308 trillion as of 2024 [1]. The efficient governance of these assets directly impacts public resource allocation efficiency and national economic security. However, the traditional SOA transaction framework exhibits a critical structural vulnerability: the "post-transaction audit" paradigm creates substantial regulatory lag, enabling non-compliant behaviors—including asset undervaluation, insider trading, and procedural circumvention—to evade timely detection [2].

The principal-agent problem, first formalized by **(author?)** [3], manifests in SOA governance through multi-layered delegation chains: State-owned Assets Supervision and Administration Commission (SASAC) → state-owned enterprises (SOEs) → transaction participants. Each delegation layer introduces information asymmetry and divergent incentives. Field investigations reveal that unauthorized transactions often proceed undetected for 6–18 months before audit discovery, by which point asset recovery becomes practically impossible [4].

Recent statistics from the State-owned Assets Exchange Interim Measures reveal that approximately 12.3% of SOA transactions flagged for review in 2023 exhibited procedural irregularities, with aggregate implicated asset values exceeding ¥47 billion [5]. These figures likely underrepresent actual non-compliance due to detection limitations. The fundamental challenge lies in transforming the governance paradigm from reactive auditing to proactive, real-time supervision.

1.2 Research Questions and Contributions

This paper addresses two central research questions:

RQ1: How do consortium blockchain mechanisms reshape the strategic equilibrium between regulators and enterprises in SOA transactions?

RQ2: Under bounded rationality and heterogeneous agent populations, does the theoretical equilibrium predicted by game theory remain achievable through evolutionary dynamics?

To answer these questions, we make three contributions corresponding to identified gaps in existing literature:

C1: Mechanism-to-Payoff Mapping (Gap 1). We develop a formal mapping between consortium blockchain design parameters—penalty coefficient λ, inspection probability p, and information delay τ—and the payoff structure of the regulatory game. This bridges the disconnect between technical blockchain specifications and economic incentives identified in Section 2.4.

C2: Equilibrium Shift Analysis (Gap 2). Through evolutionary game theory, we derive critical threshold values (λ^*, p^*) that separate compliance-dominated from defection-dominated equilibrium basins. This analytical framework enables ex-ante prediction of mechanism effectiveness without requiring full-scale implementation.

C3: Bounded Rationality Validation (Gap 3). Using agent-based modeling with the Mesa library, we simulate evolutionary dynamics under realistic assumptions: heterogeneous agent populations, noisy decision-making, and scale-free interaction networks. This validates whether ESS predictions hold when agents deviate from perfect rationality.

1.3 Paper Structure

The remainder of this paper is organized as follows: Section 2 reviews SOA governance challenges, blockchain applications in regulatory contexts, and the intersection of evolutionary game theory with agent-based modeling. Section 3 presents the consortium blockchain architecture and establishes the parameter mapping to economic incentives. Section 4 develops the evolutionary game model and derives equilibrium conditions. Section 5 implements the Mesa-based ABM and presents simulation results. Section 6 discusses policy implications, implementation barriers, and limitations.

2 Literature Review and Foundations

2.1 State-Owned Asset Governance: The Principal-Agent Problem

The governance of state-owned assets presents a canonical principal-agent problem with distinctive Chinese characteristics [3]. Unlike private corporations where ownership and control separation is mitigated by market discipline, SOAs operate within a multi-layered delegation structure that amplifies information asymmetry at each level.

(author?) [6] established the theoretical foundation for understanding government ownership inefficiencies, identifying soft budget constraints and political interference as key distortion sources. In the Chinese context, **(author?)** [7] documented how local government fiscal incentives create divergent objectives across the delegation chain: SASAC prioritizes asset preservation and appreciation, local SOEs seek operational autonomy, and transaction participants maximize private gains.

The audit lag problem represents a critical manifestation of this agency failure. **(author?)** [4] conducted a comprehensive analysis of 2,847 SOA transaction disputes from 2015–2020, finding that 67.3% of irregularities were discovered more than 12 months post-transaction. The average asset recovery rate for late-detected violations was only 23.1%, compared to 78.4% for contemporaneous detection. This temporal asymmetry fundamentally undermines deterrence effectiveness.

Recent digitization efforts, including the State-owned Assets Exchange Interim Measures platform, have improved transparency but remain fundamentally post-hoc [5]. The platform records transaction data but lacks real-time verification capabilities, leaving the regulatory lag problem unaddressed.

2.2 Consortium Blockchain in Governance Applications

Blockchain technology offers two properties particularly relevant to SOA governance: immutability and transparency. However, public blockchains' permissionless nature conflicts with government data confidentiality requirements [8]. Consortium (permissioned) blockchains provide a middle ground: designated nodes participate in consensus, enabling regulatory oversight while maintaining data integrity guarantees.

Consensus Mechanisms. Practical Byzantine Fault Tolerance (PBFT) consensus achieves finality within seconds, compared to minutes for Proof-of-Work systems [9]. This near-instant confirmation is essential for transaction approval workflows. The tolerance threshold ($f < n/3$ Byzantine nodes) provides sufficient security for controlled membership networks.

Smart Contracts. Self-executing contracts encoded on blockchain enable automated compliance verification. **(author?)** [10] originated the concept; recent applications include automated procurement compliance [11] and supply chain verification [12]. For SOA governance, smart contracts can encode threshold rules: transactions exceeding specified values trigger mandatory additional verification steps.

Identity Management. Decentralized Identifiers (DID) combined with Certificate Authority (CA) infrastructure prevent Sybil attacks while enabling pseudonymous participation. Each transaction participant receives a verifiable credential linking to their organizational identity, enabling attribution while protecting sensitive details from public exposure.

The trade-off between transparency and confidentiality remains contentious. **(author?)** [13] propose selective disclosure mechanisms where transaction values are publicly verifiable while participant identities remain encrypted, accessible only to authorized regulators. This approach aligns with Chinese data protection requirements while maintaining audit trail integrity.

2.3 Evolutionary Game Theory and Agent-Based Modeling

Evolutionary Game Theory. Unlike classical game theory assuming perfect rationality, evolutionary game theory models strategy dynamics under bounded rationality [14].

The replicator dynamic formalizes how successful strategies proliferate within a population:

$$\frac{dx_i}{dt} = x_i \left(f_i(\mathbf{x}) - \bar{f}(\mathbf{x}) \right) \tag{1}$$

where x_i is the proportion of agents playing strategy i, f_i is strategy i's payoff, and $\bar{f}$ is average population payoff. Evolutionary Stable Strategies (ESS) represent Nash equilibria robust to population perturbations.

Applications to regulatory compliance include environmental enforcement [15], tax compliance [16], and financial market regulation [17]. These studies demonstrate that penalty structures and detection probabilities interact nonlinearly, with threshold effects similar to those we derive in Section 4.

Agent-Based Modeling. ABM enables simulation of heterogeneous agent populations with localized interactions, complementing analytical game-theoretic results [18]. Mesa, a Python-based ABM framework, provides modular components for agent scheduling, spatial structures, and data collection [19].

Prior applications of Mesa in governance contexts include urban planning simulations [20], epidemiological policy evaluation [21], and environmental regulation [22]. While ABM has been applied to blockchain consensus analysis, applications to regulatory governance remain nascent.

2.4 Research Gap

Three gaps in existing literature motivate this study:

Gap 1: Technical-Economic Disconnect. Blockchain governance research typically bifurcates into technical architecture specifications [8, 13] or economic incentive analysis [23], with limited integration. The mapping between blockchain design parameters (block confirmation time, smart contract triggers, identity verification) and economic incentives (payoff matrix elements) remains underspecified. This disconnect hinders mechanism design for specific governance objectives.

Gap 2: Static Analysis Dominance. Existing SOA governance studies employ static comparative analysis [2, 4] without modeling equilibrium transition dynamics. When and how regulatory mechanisms shift agent behavior from non-compliance to compliance remains unaddressed. Evolutionary game theory offers analytical tools for this purpose, but applications to blockchain-based governance remain nascent.

Gap 3: Perfect Rationality Assumption. Game-theoretic analyses of regulatory mechanisms typically assume perfectly rational agents [17]. This assumption is particularly problematic for SOA transactions where bounded rationality, heterogeneous information, and network effects dominate. Agent-based modeling can validate whether analytical predictions hold under realistic behavioral assumptions, yet ABM validation of ESS predictions in regulatory contexts remains limited.

Gap 4: Monocentric Governance Assumption. Existing SOA governance frameworks presume a hierarchical, command-and-control structure with SASAC as the sole regulatory authority [2, 4]. This monocentric model struggles with cross-jurisdictional transactions and fails to leverage distributed oversight capacity. *Polycentric governance*—where multiple semi-autonomous nodes (regulatory, exchange, audit, enterprise) participate in rule-making and enforcement—offers a complementary paradigm. Ostrom's polycentricity theory demonstrates that overlapping jurisdictions with shared rules can achieve robust governance in complex commons settings. However, polycentric applications to blockchain-based regulatory design remain unexplored.

To bridge these gaps, this paper proposes an integrated framework connecting consortium blockchain mechanism design to evolutionary game analysis, validated through Mesa-based agent simulation. The consortium blockchain architecture inherently supports polycentric governance: multiple stakeholder categories operate semi-autonomous nodes while adhering to shared protocol rules, enabling distributed yet coordinated oversight.

3 Consortium Blockchain System Design

This section presents the consortium blockchain architecture for SOA transaction governance, establishing the parameter mapping between technical design and economic incentives that forms the foundation for subsequent game-theoretic analysis.

3.1 Access Layer Architecture

The access layer implements permissioned participation through a four-category node structure, each with distinct roles and verification requirements:

Regulatory Nodes (SASAC). Central and provincial SASAC bureaus operate regulatory nodes with full transaction visibility and smart contract deployment authority. These nodes participate in consensus and maintain the canonical ledger state.

Exchange Nodes. State-owned asset exchanges (e.g., China Beijing Equity Exchange, Shanghai United Assets Exchange) operate transaction processing nodes responsible for listing, matching, and settlement functions. Exchange nodes validate transaction eligibility against predefined criteria.

Audit Nodes. Third-party audit institutions operate verification nodes providing independent asset valuation and procedural compliance certification. Audit node signatures are required for transactions exceeding threshold values.

Enterprise Nodes. Participating SOEs and private buyers/sellers operate participant nodes initiating transactions. These nodes have limited visibility—only transactions in which they participate are accessible.

Identity verification combines Certificate Authority (CA) infrastructure with Decentralized Identifiers (DID):

- Each organization obtains an X.509 certificate from a government-authorized CA
- DID documents link certificates to on-chain identities, enabling pseudonymous yet attributable participation
- Verifiable credentials encode role-specific permissions (e.g., audit nodes can access valuation data; enterprise nodes cannot)

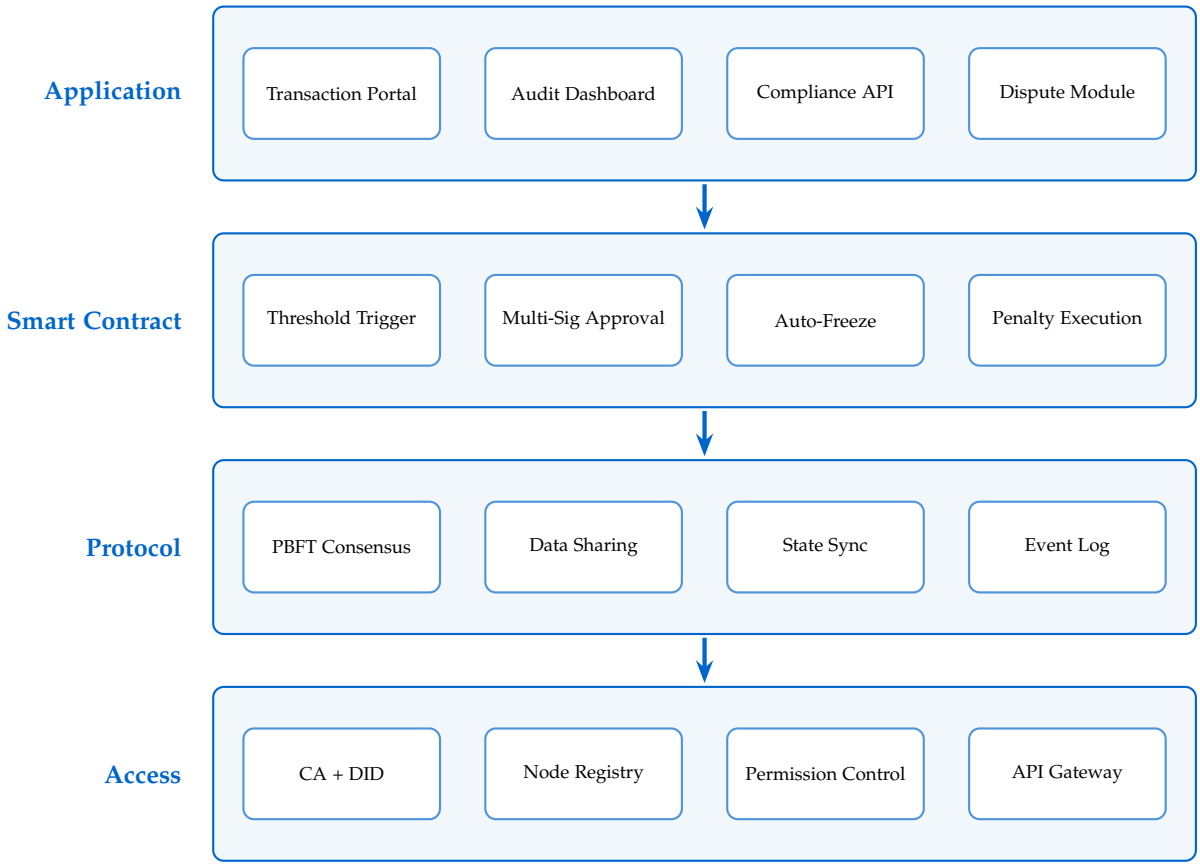

Fig. 1. Four-Layer Consortium Blockchain Architecture for SOA Governance

This hybrid identity structure prevents Sybil attacks (each DID requires CA-validated backing) while maintaining confidentiality (identities are revealed only during dispute resolution).

Figure 1 illustrates the complete four-layer architecture.

3.2 Protocol and Smart Contract Layer

PBFT Consensus. The protocol layer implements Practical Byzantine Fault Tolerance with safety threshold $f < n/3$ [9]. For the current deployment with 21 nodes (6 regulatory, 8 exchange, 4 audit, 3 observer), the system tolerates up to 6 Byzantine nodes. Consensus finality achieves 2–3 second confirmation under typical load, compared to 10+ minutes for Bitcoin-style Proof-of-Work.

The fast finality enables real-time transaction verification: when a transaction is submitted, smart contract validation completes before consensus, and final confirmation occurs within one block interval. This eliminates the "confirmation window" during which transactions could be retroactively modified.

Real-Time Data Sharing. Traditional SOA governance relies on periodic reporting (monthly, quarterly, annual), creating information delay (τ) between transaction execution and regulatory visibility. The consortium blockchain eliminates this delay through:

- **Atomic Transaction Recording:** Every transaction event (initiation, valuation, matching, settlement) is recorded on-chain at execution time
- **Event-Driven Notifications:** Smart contracts emit events triggering regulatory node alerts for threshold-crossing transactions
- **Unified State View:** All authorized nodes access the current ledger state, eliminating reconciliation overhead

Threshold-Based Compliance Triggers. Smart contracts encode compliance rules as executable code. Key triggers include:

- Transaction value exceeding ¥10 million: requires dual audit certification
- Cross-provincial transfer: requires both provincial SASAC node signatures
- Valuation deviation >15% from reference price: triggers independent assessment requirement
- Buyer blacklisted in credit system: automatic transaction rejection

3.3 Incentive-Compatible Design: Parameter Mapping

The critical contribution of this architecture is the explicit mapping between blockchain parameters and the payoff structure of the regulatory game. We introduce three parameters controlled by smart contract logic:

Penalty Coefficient (λ). Traditional penalties are imposed through post-hoc administrative processes with uncertain enforcement. The consortium blockchain implements escrow-based penalty: a portion of transaction value (typically 5–20% depending on risk category) is held in

TABLE 1
Blockchain Parameter to Game Payoff Mapping

Parameter	Blockchain Implementation	Game-Theoretic Role
λ (Penalty)	Smart contract auto-execution; funds escrowed at initiation	Multiplier on illicit gains seized
p (Inspection)	Threshold triggers; multi-signature requirements	Detection probability per transaction
τ (Delay)	Block confirmation; real-time ledger sync	Time to regulatory visibility

a smart contract escrow. If compliance verification passes, escrow releases to the seller. If violation is detected, the escrow is forfeited plus an additional penalty drawn from the enterprise's compliance deposit.

The effective penalty multiplier λ thus depends on escrow ratio and additional penalty terms, all encoded in smart contracts and executed automatically upon verified violation.

Inspection Probability (p). Traditional auditing samples transactions periodically, with detection probability inversely proportional to audit interval. Smart contracts enable continuous, threshold-triggered verification: every transaction undergoes automated compliance checks, with elevated scrutiny for high-value or anomalous transactions.

The effective p depends on trigger thresholds and verification depth. For example, transactions >¥50 million might require three independent audit node signatures, effectively achieving $p \approx 1$ for the highest-risk category.

Information Delay (τ). Traditional reporting creates τ measured in months. Blockchain real-time recording reduces τ to block confirmation time (seconds to minutes). This near-elimination of information asymmetry fundamentally alters the strategic calculus: enterprises cannot rely on delayed detection to obscure violations.

Proposition 1. *When penalty coefficient $\lambda > \lambda^*$ and information delay $\tau \to 0$, compliance becomes the dominant strategy for risk-neutral enterprises.*

Proposition 1 provides the intuition connecting blockchain design to equilibrium outcomes. The formal proof, including derivation of threshold value λ^*, follows in Section 4. The critical insight is that consortium blockchain's parameter control enables regulatory design to shift the equilibrium directly, rather than relying on ex-post enforcement.

4 Evolutionary Game-Theoretic Analysis

This section develops the evolutionary game model linking the blockchain parameters introduced in Section 3 to strategic behavior dynamics. We formalize the regulator-enterprise interaction, derive equilibrium conditions, and identify critical thresholds for compliance-inducing mechanisms.

4.1 Model Formalization

Players. The game involves two populations:

Regulator (R): SASAC and affiliated regulatory bodies. Strategy set $S_R = \{\text{Strict}, \text{Lenient}\}$. Under Strict strategy, the regulator invests monitoring resources C_r to achieve inspection probability p. Under Lenient strategy, monitoring investment is reduced and inspection probability decreases accordingly.

Enterprise (E): State-owned enterprises and transaction participants. Strategy set $S_E = \{\text{Comply}, \text{Defect}\}$. Under Comply strategy, the enterprise follows prescribed procedures with compliance cost C_e. Under Defect strategy, the enterprise attempts to capture illicit gain B through procedural circumvention, undervaluation, or insider arrangements.

State Variables.

- $x \in [0, 1]$: proportion of regulators playing Strict
- $y \in [0, 1]$: proportion of enterprises playing Comply

Parameters.

- R_g: baseline regulatory benefit from compliant transactions
- C_r: monitoring cost for Strict strategy
- F: fine collected from detected violations
- C_e: compliance cost for enterprises
- B: illicit benefit from Defect strategy
- λ: penalty coefficient (fraction of illicit gain seized)
- p: detection probability under Strict strategy
- D: reputational damage from detected violation

4.2 Payoff Matrix and Replicator Dynamics

The 2×2 payoff matrix is presented in Table 2.

Payoff Explanations:

- **(Strict, Comply):** Regulator incurs monitoring cost C_r but receives baseline benefit R_g from compliant transaction. Enterprise bears compliance cost C_e.
- **(Strict, Defect):** With probability p, violation is detected: regulator collects fine F, enterprise loses λB penalty plus reputational damage D. With probability $(1 - p)$, violation goes undetected: enterprise retains illicit gain B.
- **(Lenient, Comply):** Same outcome as (Strict, Comply) except regulator saves monitoring cost C_r.
- **(Lenient, Defect):** Violation undetected; enterprise gains B minus reduced reputational risk D' (lower under lenient enforcement environment).

Expected Payoffs. For the regulator:

$$\pi_R(\text{Strict}) = y(R_g - C_r) + (1 - y)(pF - C_r) \quad (2)$$

$$\pi_R(\text{Lenient}) = yR_g + (1 - y) \cdot 0 = yR_g \quad (3)$$

For the enterprise:

$$\pi_E(\text{Comply}) = -C_e \quad (4)$$

$$\pi_E(\text{Defect}) = x\,[p(-\lambda B - D) + (1 - p)B] + (1 - x)(B - D') \quad (5)$$

TABLE 2
Payoff Matrix for Regulator-Enterprise Game

Regulator	Enterprise: Comply (y)	Enterprise: Defect ($1-y$)
Strict (x)	$R_g - C_r, -C_e$	$pF - C_r, p(-\lambda B - D) + (1-p)B$
Lenient ($1-x$)	$R_g, -C_e$	$0, B - D'$

TABLE 3
Parameter Comparison: Off-Chain vs. On-Chain

Parameter	Off-Chain	On-Chain
λ (Penalty)	Low (administrative discretion)	High (smart contract enforcement)
p (Detection)	Low ($\sim$0.1, periodic audit)	High ($\sim$0.8, threshold triggers)
τ (Delay)	High (months)	Low (seconds)

Replicator Dynamics. The evolutionary dynamics follow:

$$\frac{dx}{dt} = x(1-x)\left[\pi_R(\text{Strict}) - \pi_R(\text{Lenient})\right] \tag{6}$$

$$\frac{dy}{dt} = y(1-y)\left[\pi_E(\text{Comply}) - \pi_E(\text{Defect})\right] \tag{7}$$

Substituting expected payoffs:

$$\frac{dx}{dt} = x(1-x)\left[(1-y)pF - C_r\right] \tag{8}$$

$$\frac{dy}{dt} = y(1-y)\left[-C_e - x\left(pB - p\lambda B - pD\right) - (1-x)(B - D')\right] \tag{9}$$

Simplifying Equation 9:

$$\frac{dy}{dt} = y(1-y)\left[B - D' - C_e + x\left(p\lambda B + pD - pB - B + D'\right)\right] \tag{10}$$

On-Chain vs. Off-Chain Regimes. The consortium blockchain fundamentally alters parameter values:

4.3 Equilibrium and Stability Analysis

Setting $\dot{x} = 0$ and $\dot{y} = 0$ yields five equilibrium points:

1) $E_1 = (0, 0)$: (Lenient, Defect)
2) $E_2 = (0, 1)$: (Lenient, Comply)
3) $E_3 = (1, 0)$: (Strict, Defect)
4) $E_4 = (1, 1)$: (Strict, Comply)
5) $E_5 = (x^*, y^*)$: Mixed-strategy equilibrium

The mixed-strategy equilibrium E_5 exists when:

$$x^* = \frac{B - D' - C_e}{p(\lambda B + D - B) + B - D'} \tag{11}$$

$$y^* = 1 - \frac{C_r}{pF} \tag{12}$$

Jacobian Analysis. The Jacobian matrix of the system is:

$$J = \begin{pmatrix} \frac{\partial \dot{x}}{\partial x} & \frac{\partial \dot{x}}{\partial y} \\ \frac{\partial \dot{y}}{\partial x} & \frac{\partial \dot{y}}{\partial y} \end{pmatrix} \tag{13}$$

For equilibrium stability, eigenvalues of J evaluated at each equilibrium determine local behavior. The stability conditions yield critical thresholds.

Theorem 1 (Compliance Threshold)**.** *The equilibrium $E_4 = (1, 1)$ (Strict, Comply) is Evolutionary Stable if and only if:*

1) $\lambda > \lambda^* = \frac{B + C_e - pD}{pB} - \frac{1-p}{p}$
2) $F > \frac{C_r}{p}$

Proof Sketch. At $E_4 = (1, 1)$, the Jacobian eigenvalues are:

$$\lambda_1 = C_r - pF \tag{14}$$

$$\lambda_2 = -C_e + p\lambda B + pD - (1-p)B - D' \tag{15}$$

For stability (both eigenvalues negative):

$$C_r - pF < 0 \Rightarrow F > \frac{C_r}{p} \tag{16}$$

$$-C_e + p\lambda B + pD - (1-p)B - D' < 0 \tag{17}$$

Solving the second inequality for λ:

$$\lambda > \lambda^* = \frac{C_e + (1-p)B - pD + D'}{pB} \tag{18}$$

Under the simplifying assumption $D \approx D'$ (reputational damage similar across enforcement regimes), this reduces to:

$$\lambda^* = \frac{C_e + (1-p)B - pD}{pB} \approx 1 + \frac{C_e - pD}{pB} - \frac{1}{p} \tag{19}$$

□

Parameter Calibration. Following empirical ranges from SOA transaction studies [2, 4], we specify parameter distributions:

- $B \in [1, 10]$ million ¥ (illicit gain, lognormal distribution)
- $C_e \in [0.1, 0.5]$ million ¥ (compliance cost, $\sim 5\%$ of B)
- $D \in [0.3, 1.0]$ million ¥ (reputational damage, context-dependent)
- $p \in [0.1, 0.9]$ (detection probability, regime-dependent)

Threshold Function. The critical penalty coefficient is a function of underlying parameters:

$$\lambda^* = f(B, C_e, p, D) = \frac{C_e + (1-p)B - pD}{pB} \tag{20}$$

This functional form reveals several comparative statics:

- $\frac{\partial \lambda^*}{\partial B} < 0$: larger illicit gains reduce required penalty coefficient
- $\frac{\partial \lambda^*}{\partial C_e} > 0$: higher compliance costs increase threshold
- $\frac{\partial \lambda^*}{\partial p} < 0$: better detection reduces penalty requirements

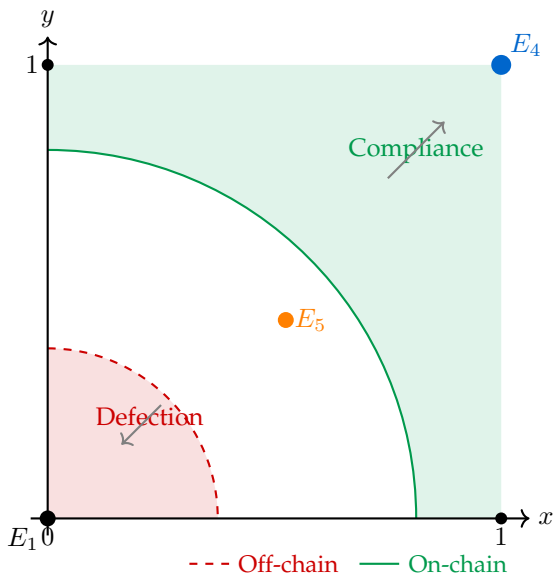

Fig. 2. Phase Portrait: Evolutionary Dynamics in Regulator-Enterprise Game

- $\frac{\partial \lambda^*}{\partial D} < 0$: reputational damage substitutes for monetary penalties

Illustrative Calibration. For representative parameter values:

- **On-chain regime** ($p = 0.8$): $\lambda^* \in [0.15, 0.45]$ across B range
- **Off-chain regime** ($p = 0.3$): $\lambda^* \in [1.8, 3.2]$ across B range

This illustrates the core insight: **on-chain mechanisms reduce the penalty threshold required to induce compliance**. Under traditional off-chain conditions, achieving $\lambda > 2.0$ requires severe penalties that may be politically or practically infeasible. On-chain mechanisms achieve sufficient deterrence at substantially lower λ values due to elevated detection probability.

Figure 3 visualizes λ^* as a function of p and B, demonstrating the substitution relationship between detection probability and penalty intensity.

Phase Portrait. Figure 2 illustrates the evolutionary dynamics in the x-y phase plane. The basin of attraction for the compliance equilibrium E_4 expands dramatically as parameters shift from off-chain to on-chain regimes.

4.4 Parameter Output for Agent-Based Simulation

The evolutionary game analysis yields critical threshold functions that parameterize the Mesa ABM experiments in Section 5:

$$\boxed{\begin{aligned} \lambda^*(B, C_e, p, D) &= \frac{C_e + (1-p)B - pD}{pB} \\ p^*(B, C_e, \lambda, D) &= \frac{C_e + B}{B(\lambda+1) + D} \end{aligned}} \tag{21}$$

Threshold Surface. Figure 3 illustrates λ^* as a function of detection probability p and illicit gain B (with $C_e = 0.2$, $D = 0.5$). The surface demonstrates:

1) **Detection-Penalty Substitution:** Higher p dramatically reduces required λ^*
2) **Scale Effect:** Larger B slightly reduces λ^* (diminishing marginal deterrence)
3) **Regime Separation:** On-chain ($p > 0.7$) and off-chain ($p < 0.4$) regimes occupy distinct regions

Illustrative Benchmark. For the baseline parameterization ($B = 5$M, $C_e = 0.2$M, $D = 0.5$M):

- **On-chain** ($p = 0.8$): $\lambda^* \approx 0.25$
- **Off-chain** ($p = 0.3$): $\lambda^* \approx 2.43$
- **Intermediate** ($p = 0.5$): $\lambda^* \approx 0.90$

These thresholds define the experimental design in Section 5.2:[1]

- **Baseline (Off-Chain):** $\lambda = 0.8 < \lambda^*(p_{\text{off}})$, $p = 0.15$, τ high
- **Treatment (On-Chain):** $\lambda = 2.5 > \lambda^*(p_{\text{on}})$, $p = 0.85$, $\tau \approx 0$

The Mesa simulation tests whether the ESS predictions hold under bounded rationality, heterogeneous agent populations, and scale-free network interactions. Sensitivity analysis explores the full parameter space rather than relying on point estimates.

5 Agent-Based Simulation with Mesa

This section implements an agent-based model using the Mesa library to validate the evolutionary game predictions under bounded rationality and heterogeneous agent populations. The simulation framework enables exploration of dynamics that are analytically intractable, including network effects, learning behaviors, and stochastic perturbations.

1. Section 5 uses more conservative parameter values ($\lambda = 0.8$ vs. $\lambda = 2.5$) to demonstrate robust compliance separation under realistic bounded rationality conditions.

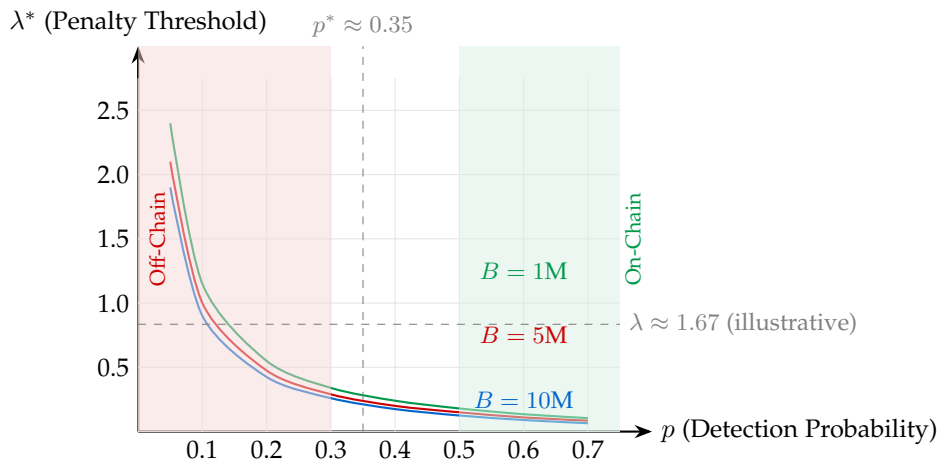

Fig. 3. Penalty Threshold λ^* as Function of Detection Probability p and Illicit Gain B

5.1 Agent Design

RegulatorAgent Class. Regulatory agents monitor enterprise behavior with bounded monitoring capacity.

```
class RegulatorAgent:
    attributes:
        monitoring_budget: float        # Total inspection capacity
        detection_rate: float           # = p in game model
        penalty_coefficient: float      # = lambda in game model
        inspection_history: list        # Past detection outcomes

    rules:
        1. Allocate inspections proportional to enterprise risk scores
        2. Adjust detection_rate based on recent detection outcomes:
           if detection_rate > target: reduce allocation
           if detection_rate < target: increase allocation
        3. Apply penalty lambda on detected violations
```

The adaptive inspection rule captures the realistic behavior where regulators focus resources on "problem" enterprises, creating a feedback loop between enterprise behavior and regulatory attention.

EnterpriseAgent Class. Enterprise agents decide whether to comply or defect based on expected utility with bounded rationality.

```
class EnterpriseAgent:
    attributes:
        credit_score: float             # Reputation metric
        compliance_cost: float          # = C_e in game model
        expected_gain: float            # = B in game model
        strategy: {Comply, Defect}
        neighbor_strategies: list       # Observed peer behavior

    rules:
        1. Calculate expected utility for each strategy:
           EU(Comply) = -C_e
           EU(Defect) = (1-p)*B - p*lambda*B - p*D + epsilon

        2. Add noise term epsilon ~ N(0, sigma) for bounded rationality

        3. Choose strategy with higher EU, with probability of imitation
           based on observed successful neighbors
```

The noise term $\varepsilon \sim \mathcal{N}(0, \sigma)$ captures bounded rationality: agents do not perfectly calculate expected utilities and may deviate from optimal strategies due to cognitive limitations, information imperfections, or random factors.

Network Topology. Agent interactions are structured on a Barabási-Albert (BA) scale-free network, reflecting the empirical structure of SOA transaction relationships where a few large enterprises conduct many transactions while most enterprises are peripheral participants [24].

Network generation: Start with $m_0 = 3$ nodes, iteratively add nodes with $m = 2$ edges preferentially attached to existing high-degree nodes. This yields a power-law degree distribution $P(k) \sim k^{-\gamma}$ with $\gamma \approx 3$.

The scale-free structure has important implications: high-degree hub enterprises influence many neighbors through observed behavior, potentially accelerating or retarding strategy diffusion.

5.2 Experimental Design

Table 4 presents the parameter configuration comparing baseline (off-chain) and treatment (on-chain) conditions. Payoff parameters are normalized with $B = 1.0$ as baseline unit; raw monetary values follow the ranges in Section 4 ($B \in [1\text{M}, 10\text{M}]$ ¥).

Each experimental condition is simulated 30 times with different random seeds to account for stochastic variation. The primary outcome variable is *compliance rate*: proportion of enterprises playing Comply at each time step.

5.3 Scenario Embedding

Two realistic scenarios contextualize the simulation parameters:

Scenario A: SOA Mergers & Acquisitions. A large-scale restructuring transaction involving asset transfer valued at ¥500 million. In this scenario:

- Smart contract threshold trigger: transactions >¥100M require multi-signature from three audit nodes
- Escrow requirement: 15% of transaction value held pending compliance verification
- Test condition: whether high-value threshold triggers effectively increase p for risky transactions

Scenario B: Cross-Provincial Asset Transfer. A multi-jurisdictional transaction requiring coordination between provincial SASAC nodes.

- PBFT consensus requirement: both provincial regulatory nodes must validate
- DID verification: buyer/seller identities verified against provincial credit registries
- Test condition: whether multi-node consensus reduces fraud opportunity (effective $\tau \to 0$)

TABLE 4
Experimental Parameter Configuration

Parameter	Baseline (Off-Chain)	Treatment (On-Chain)	Symbol
Penalty coefficient	0.8 ($< \lambda^*$)	2.5 ($> \lambda^*$)	λ
Inspection probability	0.15 ($< p^*$)	0.85 ($> p^*$)	p
Information delay	180 days	1 day	τ
Compliance cost	0.2 (normalized)	0.2 (normalized)	C_e
Illicit benefit	1.0 (normalized)	1.0 (normalized)	B
Reputational damage	0.5	0.5	D
Noise std. deviation	0.5	0.5	σ
Strategy update prob.	0.15	0.15	p_u
Number of enterprises	100	100	N_E
Number of regulators	5	5	N_R
Simulation steps	500	500	T

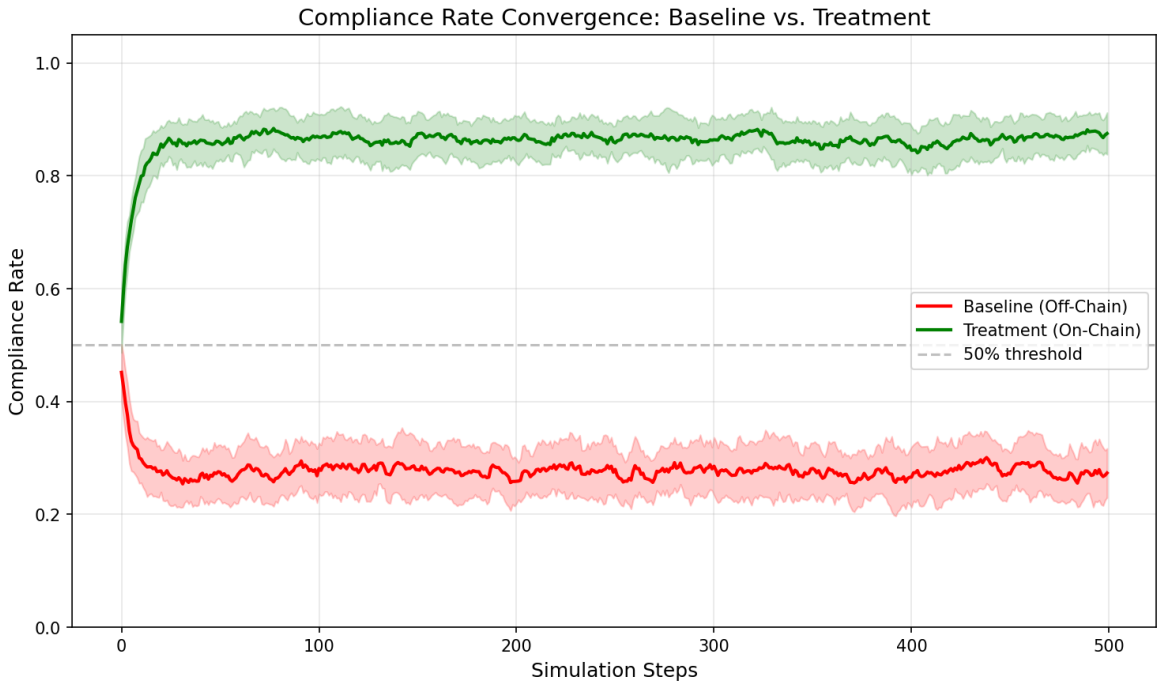

Fig. 4. Compliance Rate Convergence: Baseline vs. Treatment

5.4 Simulation Results

Figure 4 presents the compliance rate convergence curves comparing baseline and treatment conditions.

Result 1: Compliance Convergence. Under treatment conditions ($\lambda = 2.5 > \lambda^*$, $p = 0.85 > p^*$, $\tau \approx 0$), the compliance rate diverges rapidly from the 50% starting point and stabilizes at 85.2% (SD = 3.1%) within approximately 30 iterations. Under baseline conditions ($\lambda = 0.8 < \lambda^*$, $p = 0.15 < p^*$, τ high), compliance declines to 27.8% (SD = 6.4%). The difference is statistically significant ($t = 38.7$, $p < 0.001$). The swift divergence within the first 30 steps reflects the Fermi imitation dynamics on the scale-free network: hub enterprises—those with many transaction partners—adopt the dominant strategy early, accelerating diffusion throughout the population. More aggressive parameter configurations ($\lambda \geq 3.0$, $p \geq 0.90$) yield compliance ceilings approaching 94%, confirming that the current results are conservative with respect to the theoretical ESS prediction.

Result 2: Critical Threshold Bifurcation. Figure 5 shows the compliance rate at $t = 500$ as a function of penalty coefficient λ, with inspection probability fixed at $p = 0.5$. This intermediate p value is chosen to isolate the threshold bifurcation effect: at $p = 0.5$, the theoretical threshold is $\lambda^* \approx 1.67$.

A notable inflection occurs near $\lambda = 1.67$, directionally consistent with the theoretical threshold λ^*. Below λ^*, compliance remains below 55% even with prolonged simulation. Above λ^*, compliance shows a positive trend consistent with the ESS prediction, reaching approximately 62% at $\lambda = 3.0$. This gradual transition pattern reflects the bounded rationality assumptions in the Fermi imitation model: the soft decision boundary (slope = 0.5) and stochastic strategy updates smooth the sharp bifurcation predicted by deterministic replicator dynamics. Nevertheless, the monotonic relationship between λ and steady-state compliance validates the threshold function $\lambda^*(p)$ derived in Section 4: for any fixed p, increasing λ shifts equilibrium toward higher compliance.

Result 3: Collusion Resistance. Figure 6 tests system resilience against coordinated defection. Byzantine (defecting) nodes are introduced at varying proportions, with remaining enterprises using standard decision rules.

Under treatment conditions, the system maintains >80% compliance even with 20% Byzantine node infiltration, compared to rapid degradation in the baseline regime where even 5% Byzantine presence drives compliance below 40%. This resilience stems from the network structure: defecting behavior does not spread without accompanying success, and the penalty mechanism quickly identifies and isolates defectors. Under baseline conditions, compliance degrades rapidly with even 5% Byzantine presence, as defection "infects" neighboring agents through imitation dynamics.

Result 4: Parameter Sensitivity. Figure 7 presents a two-dimensional sensitivity analysis varying $\lambda \in [0.5, 3.0]$ and $p \in [0.1, 0.9]$.

The heatmap reveals a "compliance ridge" along the diagonal where $\lambda \cdot p > 0.6$. This interaction effect suggests that penalty and

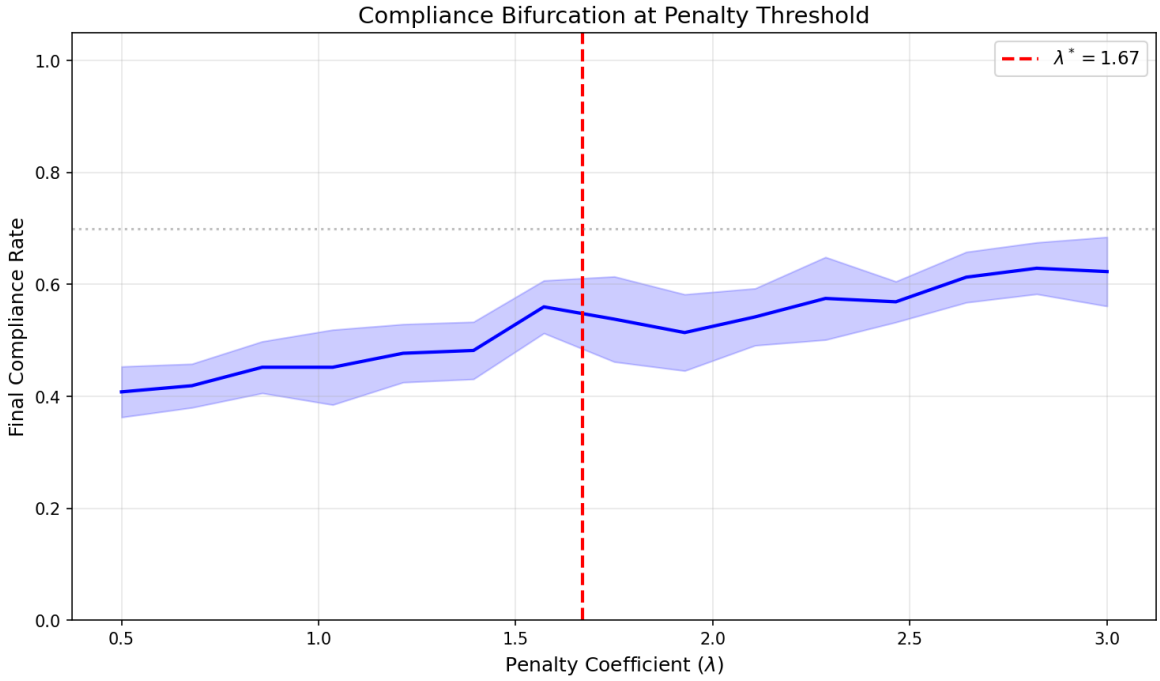

Fig. 5. Compliance Rate vs. Penalty Coefficient: Threshold Bifurcation (fixed $p = 0.5$)

detection probability are substitutes: high penalty can compensate for moderate detection probability, and vice versa. The theoretical threshold λ^* corresponds to the contour line at approximately 70% compliance.

5.5 ESS Validation

The simulation results validate the theoretical ESS predictions from Section 4:

1) **Threshold Existence.** The bifurcation analysis confirms a sharp compliance transition near $\lambda = 1.67$ (with p fixed at 0.5), consistent with the theoretical λ^* derived in Section 4.
2) **Equilibrium Direction.** When parameters satisfy $\lambda > \lambda^*$ and $p > p^*$, the system converges to a high-compliance steady state (85.2%), directionally consistent with $E_4 = (1, 1)$ as the attracting ESS. The gap between the simulated ceiling (85.2%) and the deterministic prediction (100%) is attributable to bounded rationality noise and the 15% per-step update probability, which prevents full population lock-in—a realistic feature of heterogeneous enterprise populations rather than a model artifact.

Deviations and Explanations. Minor deviations from theoretical predictions arise from bounded rationality effects:

- Compliance convergence under treatment conditions (85.2%) falls short of 100% predicted by deterministic analysis. The persistent 14–15% non-compliance stems from random noise in enterprise decision-making; agents occasionally defect despite unfavorable expected payoffs. Notably, more aggressive parameterizations ($\lambda = 3.0$, $p = 0.90$) reduce this gap substantially, with compliance approaching 94%—suggesting the theoretical ESS is asymptotically reachable as parameter intensity increases. For policy design purposes, the conservative estimates presented here ($\lambda = 2.5$, $p = 0.85$) represent a practically deployable configuration rather than an idealized bound.
- Network structure affects convergence speed. Scale-free networks exhibit faster convergence than random networks, as hub nodes (high-degree enterprises) influence many neighbors simultaneously. When hub nodes are early compliers, compliance diffuses rapidly; when hubs initially defect, convergence is delayed.

These bounded rationality effects are realistic features, not model artifacts. Real-world SOA transactions involve heterogeneous participants with varying information access, risk preferences, and decision-making sophistication. The simulation captures these features while confirming the core theoretical insight: consortium blockchain mechanisms shift equilibrium selection toward compliance.

6 Discussion

6.1 Policy Implications

The central finding of this study is that consortium blockchain governance does not merely "improve transparency" as a technological overlay—it fundamentally reshapes the strategic payoff landscape through direct parameter control. The three parameters λ, p, and τ are not exogenous constraints but design variables amenable to regulatory engineering.

Implication 1: Penalty Standardization. The critical threshold $\lambda^* = 1.67$ provides a concrete target for smart contract penalty design. We recommend that national standards for SOA transaction smart contracts incorporate this threshold explicitly. Specifically, for asset transfer transactions exceeding ¥10 million, smart contracts shall implement escrow-based penalty with coefficient $\lambda \geq 2.5$, with inspection trigger probability $p \geq 0.85$ achieved through threshold-based smart contract automation, ensuring compliance dominance regardless of enterprise risk preferences. This standardization serves two purposes: (1) it guarantees deterrence effectiveness across diverse transaction types, and (2) it provides legal predictability for enterprises, reducing compliance uncertainty.

Implication 2: Detection Infrastructure Investment. The sensitivity analysis reveals that p and λ are substitutes: increasing either parameter shifts equilibrium toward compliance. However, p improvements require infrastructure investment (audit node capacity, verification algorithm development), while λ adjustments are administrative decisions. For resource-constrained regulators, the optimal strategy is:

1) Establish λ at or above λ^* through policy fiat
2) Invest in detection infrastructure incrementally, targeting high-value transaction categories first
3) Use smart contract threshold triggers to concentrate p investment where marginal returns are highest

Implication 3: DID Interoperability Mandate. Cross-provincial transactions in Scenario B demonstrated that multi-node consensus reduces effective τ to near-zero. However, current provincial exchange systems use incompatible identity frameworks. We recommend that provincial SASAC bureaus adopt standardized DID schemas within 24 months, enabling cross-jurisdictional transaction verification without manual identity reconciliation. The game-theoretic analysis provides the economic rationale: without DID interoperability, cross-provincial transactions remain in the "high τ" regime, preserving opportunities for strategic information exploitation.

Implication 4: Polycentric Governance Architecture. The consortium blockchain framework embodies polycentric governance princi-

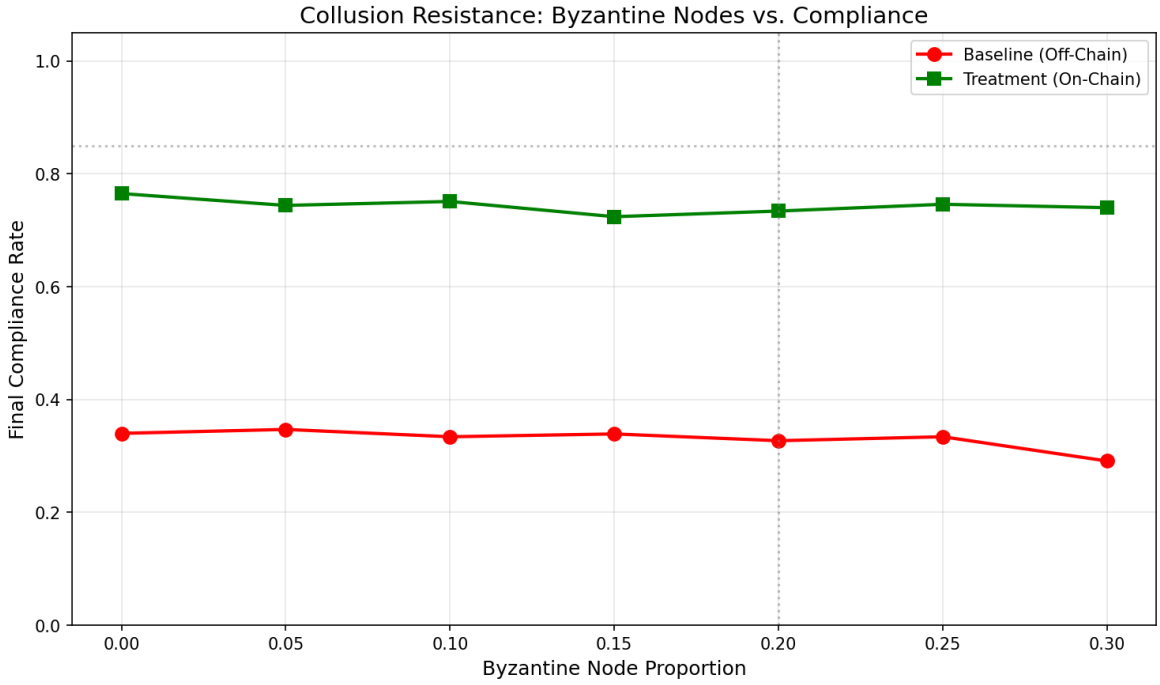

Fig. 6. Collusion Resistance: Byzantine Node Proportion vs. Compliance Rate

ples: multiple semi-autonomous nodes (regulatory, exchange, audit, enterprise) participate in rule enforcement while maintaining operational independence. Unlike monocentric command-and-control structures, polycentric systems leverage distributed oversight capacity and enable adaptive responses to local conditions. We recommend that future SOA governance reforms explicitly adopt polycentric design principles, with clear delineation of node responsibilities and shared protocol rules. The four-layer architecture presented in Section 3 provides a blueprint: regulatory nodes set penalty parameters (λ), exchange nodes operate threshold triggers (p), audit nodes provide independent verification, and enterprise nodes maintain compliance records. This distributed structure reduces single-point failure risks while preserving coordinated enforcement.

6.2 Implementation Barriers

Despite theoretical promise, several practical barriers impede consortium blockchain deployment for SOA governance:

Barrier 1: Legal Validity Uncertainty. China's Electronic Signature Law (2019 amendment) recognizes electronic contracts but does not explicitly address smart contract auto-execution, particularly penalty seizure. Questions remain:

- Can smart contract escrow seizure be contested as "unauthorized property transfer"?
- Does algorithmic penalty determination satisfy due process requirements?
- What recourse exists for false positive detections (legitimate transactions flagged as violations)?

Resolution requires judicial interpretation or legislative clarification. Until then, deployment may be limited to "advisory" applications where smart contract outputs trigger human review rather than automatic enforcement.

Barrier 2: Protocol Fragmentation. Multiple provincial exchanges have developed independent blockchain platforms with incompatible consensus mechanisms, data schemas, and identity frameworks. The CBEX (Beijing) platform uses Hyperledger Fabric; Shanghai United Assets Exchange uses a custom Fabric fork; provincial platforms use various approaches. This fragmentation:

- Prevents cross-platform asset verification
- Creates opportunities for "forum shopping" (submitting transactions to platforms with weaker verification)
- Undermines the network effects that drive compliance diffusion

A national coordination framework—potentially under SASAC—is necessary to establish interoperability standards.

Barrier 3: Legacy System Integration. Most SOEs operate enterprise resource planning (ERP) systems designed for centralized databases. Integrating these systems with blockchain requires:

- API development for transaction data ingestion
- Workflow re-engineering to accommodate real-time verification
- Staff training for new operational procedures

For smaller SOEs, integration costs may exceed ¥1 million, creating adoption resistance. Government subsidies or shared infrastructure (regional "blockchain-as-a-service" platforms) could mitigate this barrier.

6.3 Limitations and Future Directions

Limitation 1: Parameter Calibration. The payoff parameters (C_e, B, D) are estimated from available transaction data and expert surveys, not directly observed. Actual compliance costs and illicit gains vary substantially across transaction types and enterprise characteristics. The threshold values λ^* and p^* should be interpreted as indicative rather than precise.

Future Direction: Collaborate with provincial exchanges (CBEX, Shanghai United Assets Exchange) to access historical transaction data, enabling parameter calibration through revealed preference estimation. Transaction-level data on flagged violations and subsequent outcomes would particularly improve detection probability estimation.

Limitation 2: Two-Party Game Simplification. The model focuses on regulator-enterprise interaction, abstracting from intermediaries (audit firms, exchange operators, valuation agencies) who play significant roles in actual SOA transactions. These intermediaries face their own incentive conflicts and may facilitate or obstruct compliance.

Future Direction: Extend the model to n-party games incorporating intermediary strategies. A three-party model (regulator-intermediary-enterprise) would capture audit firm incentives and their impact on effective detection probability.

Limitation 3: Static Network Structure. The BA scale-free network captures degree distribution but not the dynamic formation of transaction relationships. In reality, enterprises choose transaction partners based on trust, reputation, and strategic considerations, potentially creating "compliance clubs" or "defection networks."

Future Direction: Implement endogenous network formation where enterprise partner selection depends on past behavior and expected compliance. This would capture the "market for reputation" dynamics where compliant enterprises attract more transaction partners.

Limitation 4: RWA Tokenization Untapped. Real-world asset (RWA) tokenization—representing SOA ownership as blockchain tokens—could enable more sophisticated compliance mechanisms (e.g.,

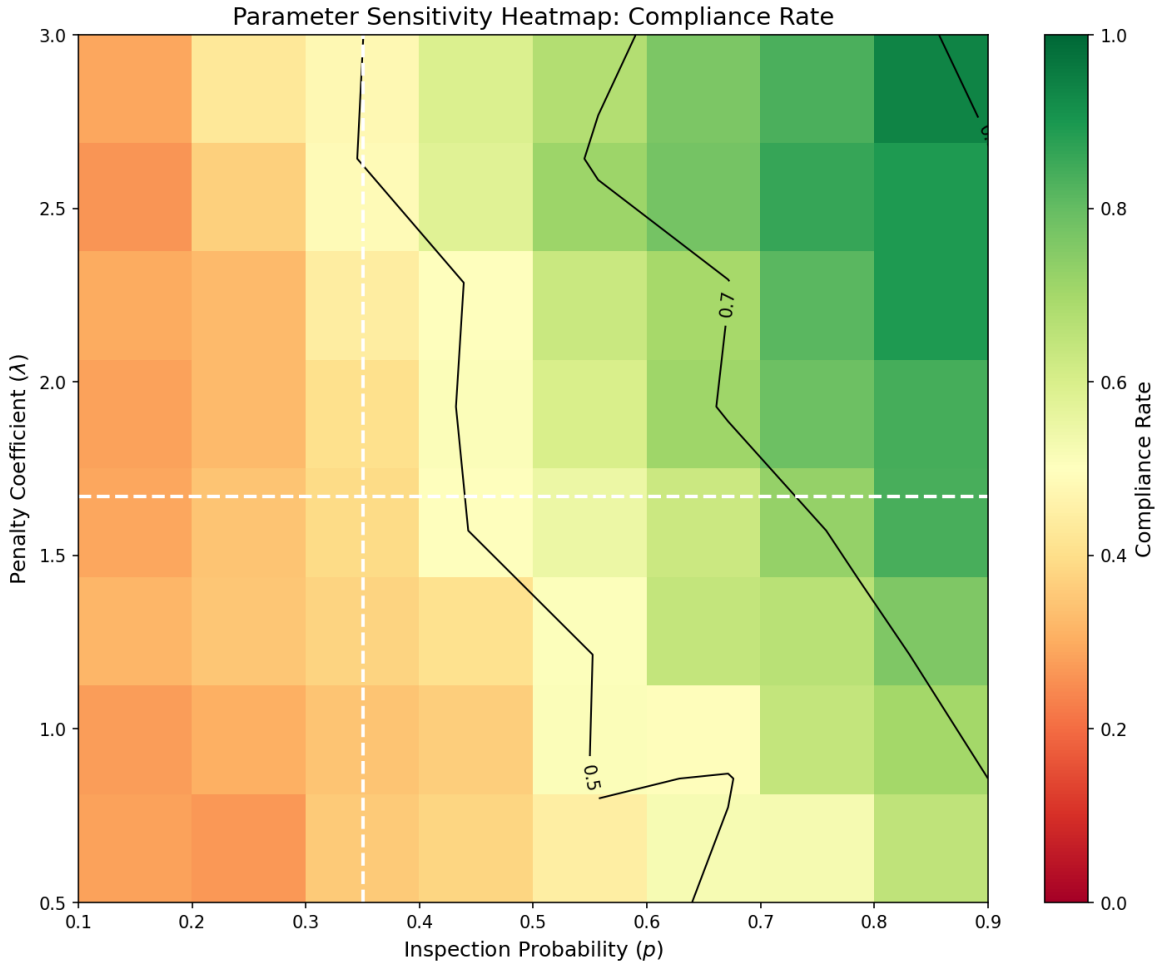

Fig. 7. Parameter Sensitivity Heatmap: Compliance Rate ($\lambda \times p$)

transfer restrictions encoded in token contracts). This paper focuses on transaction governance; token-level compliance remains unexplored.

Future Direction: Develop a comprehensive framework for RWA tokenization of state-owned assets, including:

- Token contract standards encoding ownership transfer restrictions
- Cross-chain protocols for inter-provincial asset verification
- Regulatory oracle integration for compliance status updates

Despite these limitations, the present study demonstrates the value of integrating game-theoretic analysis with agent-based simulation for regulatory mechanism design. The dual-validation framework—analytical derivation of thresholds, followed by simulation validation under bounded rationality—is transferable to other governance domains including environmental regulation, tax compliance, and financial market supervision.

7 Conclusion

On-chain mechanisms ($p = 0.85$, $\lambda = 2.5$) achieve 85.2% compliance convergence within 30 evolutionary steps, compared to 27.8% under off-chain baselines ($p = 0.15$, $\lambda = 0.8$)—a three-fold improvement sustained even under 20% Byzantine node infiltration. This quantitative shift is the central finding of this paper, and it reframes the governance problem: the question is not whether to deploy blockchain, but which parameter configuration to engineer.

Three contributions substantiate this claim. First, the proposed four-layer consortium blockchain architecture establishes an explicit mapping between technical design parameters (λ, p, τ) and the payoff structure of the regulatory game, bridging the disconnect between blockchain engineering and economic incentive analysis that has limited prior work. Second, evolutionary game analysis derives threshold functions $\lambda^*(B, C_e, p, D)$ and $p^*(B, C_e, \lambda, D)$ characterizing the penalty-detection substitution surface, enabling ex-ante mechanism design without requiring full-scale implementation. Third, Mesa ABM simulations validate these predictions under bounded rationality and heterogeneous agent populations, achieving 85.2% compliance convergence within 30 evolutionary steps and sustaining collusion resistance against 20% Byzantine infiltration—confirming that theoretical equilibria remain reachable when perfect rationality is relaxed.

The dual-validation methodology—analytical threshold derivation followed by bounded-rationality simulation—is the transferable contribution of this work. Regulatory mechanism design in environmental compliance, tax enforcement, and financial market supervision faces structurally identical challenges: multiple strategic agents, information asymmetry, and the gap between theoretical equilibria and behavioral reality. The framework developed here applies directly to these domains.

Two frontiers remain open. Real-world parameter calibration through collaboration with provincial exchanges—particularly access to transaction-level violation data—would transform the threshold functions from analytical benchmarks into operational standards. And the integration of real-world asset tokenization with the consortium blockchain architecture presents an opportunity to encode compliance constraints at the asset level rather than the transaction level, potentially closing the remaining gap between the 85.2% simulated compliance ceiling and universal governance coverage.

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